

Cybersecurity & Risk Management

Academic Report



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1. Research Five Real-World Cyber-Attacks

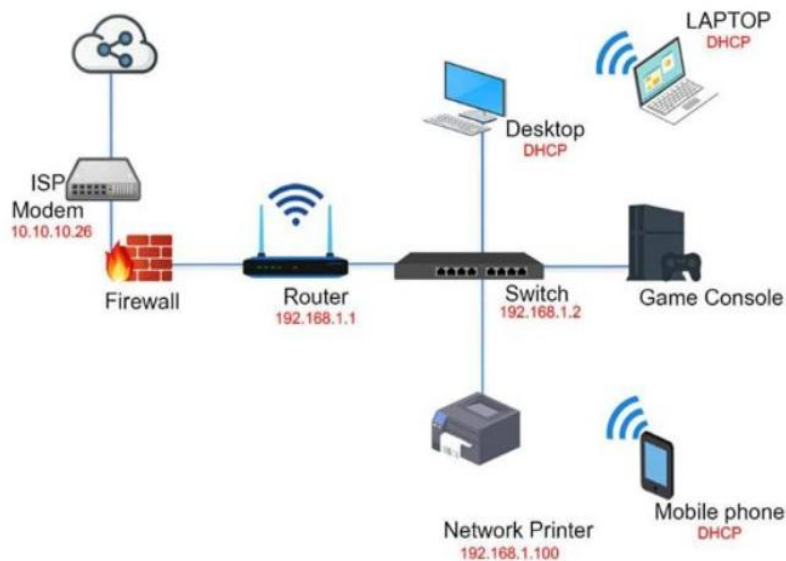
Name	Year	Organization	Type	Description	Impact
WannaCry	2017	National Health Service (NHS), UK	Ransomware	Hackers locked the computers and asked for money to unlock them.	Many hospitals could not use their computers, which delayed patient care.
Equifax Data Breach	2017	Equifax	Data Breach	Hackers stole personal information by using a weakness in the company's system.	Personal information of millions of people was leaked.
SolarWinds Attack	2020	SolarWinds	Supply Chain Attack	Hackers secretly added harmful code to a software update.	Many companies and government organizations were affected.
Colonial Pipeline Attack	2021	Colonial Pipeline	Ransomware	Hackers locked the company's computer systems with ransomware.	Fuel supply was stopped for a short time in some areas.
MOVEit Transfer Attack	2023	Organizations using MOVEit Transfer	File Transfer Exploit	Hackers used a security weakness in the software to steal data.	Data from many organizations and people was stolen.

2. CIA Triad Component(s) Violation

Cyber attack	CIA Triad Component(s)	Explanation
WannaCry	Availability	The attack locked computers, making systems and files unavailable to users.
Equifax Data Breach	Confidentiality	Hackers stole personal information that should have remained private.
SolarWinds Attack	Confidentiality, Integrity	Attackers accessed sensitive data and changed a trusted software update.
Colonial Pipeline Attack	Availability	The ransomware stopped the company's systems and disrupted fuel services.
MOVEit Transfer Attack	Confidentiality	Attackers stole sensitive data by exploiting a security weakness in the software.

3. Cybersecurity Tools Research Summary:

Tool Name	Purpose	Function	Use Case
Wireshark	To monitor and analyze network traffic.	It captures and shows the data moving through a network.	Used to find network problems and detect suspicious activity.
Nmap	To scan networks and discover devices.	It checks open ports and identifies active devices and services.	Used to find security weaknesses in a network.
Metasploit Framework	To test the security of computer systems.	It helps security professionals find and test vulnerabilities safely.	Used during penetration testing to check system security.



4. What Does a

Cybersecurity Analyst Do Every Day?

Cybersecurity analysts play a critical role in protecting an organization's systems, data, and people from constantly evolving cyber threats. Their work combines vigilance, expertise, and collaboration to keep the digital environment secure every day.

01 Monitoring and Protecting Systems

A cybersecurity analyst is responsible for protecting an organization's computers, networks, and data from cyber threats. Every day, they monitor systems for unusual activity, check security alerts, and make sure that all devices and software are working securely. They also update antivirus software, install security patches, and help keep systems protected from new threats.

02 Responding to Security Threats

Cybersecurity analysts investigate suspicious activities and respond to security incidents if they occur. They use different cybersecurity tools to find vulnerabilities, monitor network traffic, and identify possible attacks. If they discover a security issue, they work with the IT team to fix the problem quickly and prevent it from happening again. They also create reports and keep records of security events.

03 Promoting Security Awareness

Another important part of a cybersecurity analyst's job is educating employees about safe online practices. They may provide guidance on creating strong passwords, recognizing phishing emails, and protecting sensitive information. By continuously monitoring systems, responding to threats, and improving security, cybersecurity analysts help ensure that an organization's data and network remain safe and reliable.